

MD AMIR KHAN

mkhan37@stevens.edu · Stamford, CT · github.com/Mkhan2317 · linkedin.com/in/amirkhan2317 · mkhan2317.github.io

PROFESSIONAL SUMMARY

AI engineer and quantitative analyst at DXT Commodities, building production LLM systems, forecasting models, and full-stack analytics platforms used daily by the trading desk across U.S. natural gas and power markets. Sole architect on nine live systems shipped since March 2026.

EXPERIENCE

DXT Commodities North America — Stamford, CT

Applied AI Engineer & Quantitative Analyst — Market Fundamentals (LNG & Power)

March 2026 – Present

- **Pipeline transparency platform.** Built the scraping and LLM extraction pipeline that reads maintenance and capacity notices from 19 U.S. interstate gas pipelines across 8 different portal architectures (15+ format-specific scrapers covering HTML, PDF, Excel, and ASP.NET with TLS-impersonation for WAF-protected sites). Structured extraction with the Claude API turns free-form notices into typed capacity-impact records. FastAPI + SQL Server. Tracks 40+ active and 198+ upcoming restrictions on a peak day.
- **WattsApp — PJM fleet analytics.** Full-stack platform covering **4,305 units / 232.5 GW nameplate** across 2,262 plants in 13 states + DC. Overview, thermal, non-thermal, plants, insights, and natural-language query views used daily by the power desk. Next.js + React + TypeScript, FastAPI, SQL Server, Docker.
- **Wattson — multi-ISO power price analytics.** Live platform covering PJM, ERCOT, NYISO, MISO, and ISO-NE. Historical LMPs by energy/congestion/loss component, forward-contract settlements with strip aggregation, and a Price $\rightleftharpoons$ Heat-Rate toggle that connects power and gas.
- **U.S. LNG feed-gas forecasting.** Pipeline covering **126 export trains at 13 terminals (~22,600 MMcf/d nameplate)**. Three-model validation framework; **under 3% MAPE on an 8-month out-of-sample holdout**.
- **U.S. natural gas demand forecasts.** 48 models — 12 regions $\times$ 4 end-use sectors — producing daily 30-day forecasts for the Lower 48. XGBoost for weather-sensitive sectors, Ridge for the slower ones. Driven by NWS temperature forecasts and Fed industrial production data, trained on EIA state-level consumption; walk-forward cross-validation.
- **Real-time gas production model + Permian Basin intelligence.** OLS/Ridge scaling framework that combines licensed pipeline-nominations data with EIA-914 to produce a current-month U.S. production estimate inside the two-month EIA reporting lag. Permian codebase covers daily production, egress across the four major pipelines, and Waha basis pricing.
- **Pipeline Force Majeure alerts + PEPCO nodal basis screen.** Real-time alerting service polls electronic bulletin boards across **32 U.S. interstate gas pipelines** every 5 minutes and pushes Force Majeure and maintenance notices to Microsoft Teams (response time: hours $\rightarrow$ minutes). PEPCO FTR-bidding screen covers **~8.77 M hourly observations** across 61 nodes and 8+ years of PJM day-ahead data.

Stevens Institute of Technology — Hoboken, NJ

Quantitative Research Assistant — School of Business

April 2025 – February 2026

- Implemented the **ϵ -subdivision Robust PCA framework for dynamic factor portfolios** in Python with Prof. Papa Momar Ndiaye — block decomposition of the eigenspectrum against a tolerance ϵ , Gram–Schmidt construction of the closest orthonormal basis to the prior period's factors, and rupture-detection that resets factor tracking when block-mean eigenvalues shift beyond a threshold δ . Added K-Means clustering on eigenvector features as an independent signal of factor-structure change.
- Validated on ~6.5 years of daily returns across the 11 S&P 500 GICS sector ETFs, spanning the COVID-19 shock and the post-pandemic inflation cycle. Fed the robust and standard covariance estimates into a mean-variance optimizer and compared cumulative returns, 126-day rolling annualized volatility, and cluster-transition timing — the robust approach stabilized factors at portfolio volatility essentially identical to standard PCA.

EDUCATION

Stevens Institute of Technology

Hoboken, NJ

Master of Science in Financial Engineering & Analytics (STEM-designated)

North South University

Dhaka, Bangladesh

Bachelor of Business Administration · Major: Finance · Minor: Mathematics

RESEARCH

Master's Thesis — LLM-Generated Views in a Black–Litterman Portfolio

Fed repeated LLM forecasts into the Bayesian update as probabilistic views — sample mean as the view, sample variance as view uncertainty. Tested on the 50 largest S&P 500 names over Feb 2024 – Jan 2025 with biweekly rebalancing. Beat the mean-variance benchmark on risk-adjusted return with steadier allocations.

TECHNICAL SKILLS

Languages & Tools: Python, SQL, TypeScript, JavaScript · React, Next.js, FastAPI · SQL Server, Docker, AWS · Git, GitHub, GitLab

AI / LLM: LangChain · LangGraph · LangSmith · RAG · Vector Databases · LLM Agents · Prompt Engineering · Anthropic Claude & OpenAI APIs

Machine Learning: PyTorch · Scikit-learn · XGBoost · LightGBM · Pandas · NumPy · Time-Series Forecasting

Energy Markets: Natural Gas & Power · LNG Forecasting · Pipeline Capacity Analysis · ISOs / RTOs (PJM, ERCOT, NYISO, MISO, ISO-NE) · FTR Markets

CERTIFICATIONS & AWARDS

Advanced RAG · Akuna Capital Options 101 · Probability — The Science of Uncertainty and Data (MIT / edX) ·

Python and Statistics for Financial Analysis (Coursera) · FastAPI, Data Science / ML / DL / NLP Bootcamp,

Algorithmic Trading with Python, ChatGPT, ML (Udemy)

Student Member, **CFA Society New York**